OOMWOO charging-only dock, v2 (2026-10-27)
20 V brick in; DOCK+ / DOCK- pogo contacts out (energized only after the robot
is recognised by its ~130k / ~30 uF signature); 38 kHz NEC IR beacon.
CH32V003 control (Arduino core, ch32fun, PlatformIO). Refs in the 1000s: each board has its
own range (main board 1-999, dock 1000s, sensor boards 2000s-6000s).

CONTROL

File: Control.kicad_sch

POWER

File: Power.kicad_sch

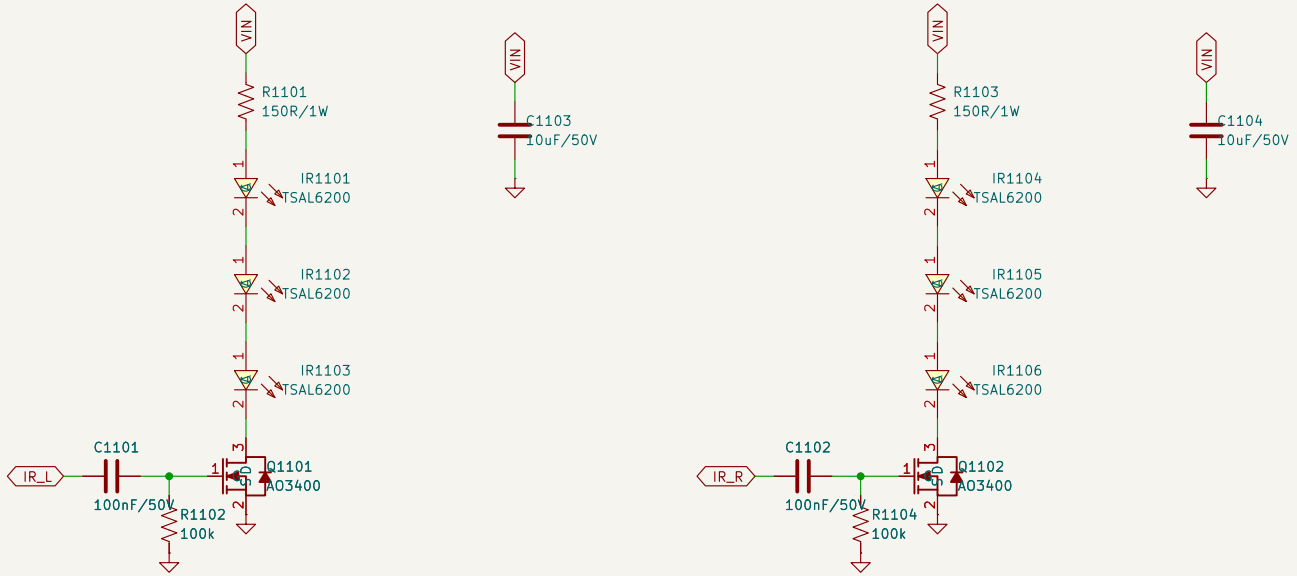
IR BEACON

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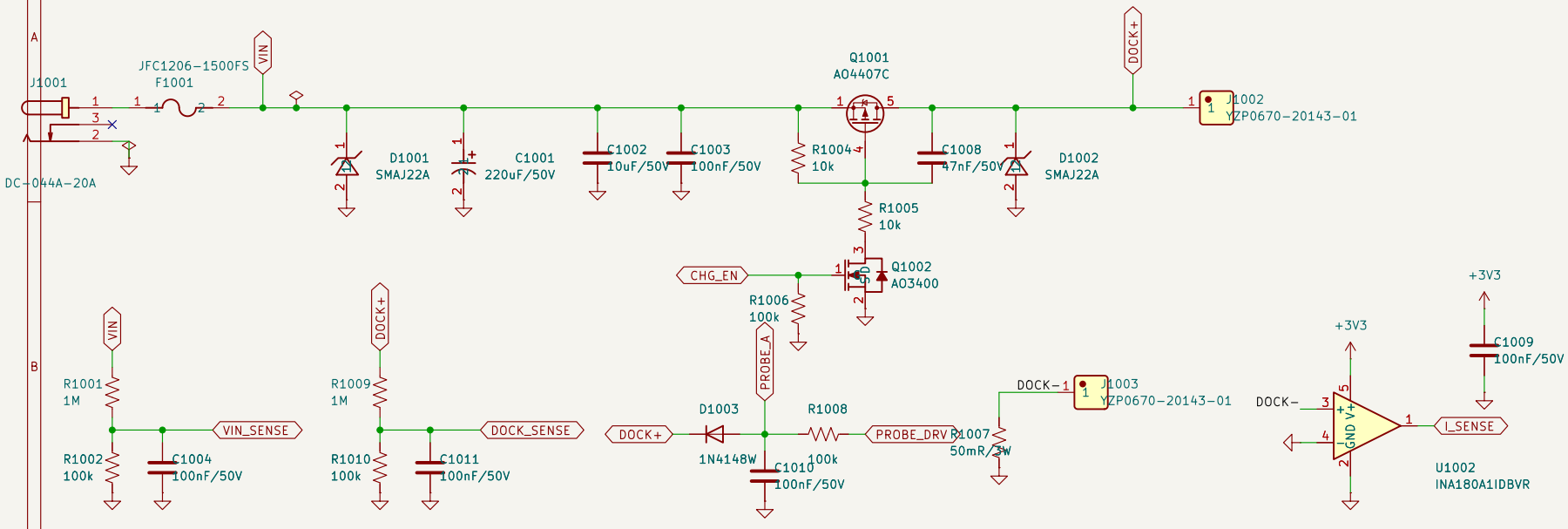

Beacon: the MCU sends 38 kHz NEC frames (TIM1 PWM, ~1/3 duty), never a steady carrier:
the robot receivers (IRM-H638T) suppress continuous 38 kHz. Left and right strings can send
different codes. Gates are AC-coupled (100 nF / 100k, ~10 ms): a pin stuck high turns the LEDs off.
Mount the three LEDs of each string fanned out (TSAL6200 is +/-17 deg) to cover the room.

**OOMWOO charging dock**

Sheet: /IR BEACON/
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***Title: IR beacon***

| Size: A4 | Date: 2026-10-27 | **Rev: 2** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 2/4 |




Robot detection (firmware): with Q1001 off, PROBE_DRV high drives ~13 uA into DOCK+.
R_robot = V(DOCK+) / ((3.3 V − V(PROBE_A)) / 100k − V(DOCK+) / 1.1M); the diode drop cancels.
v1 robot signature: ~130k (R175 + R7305 || 1.1M sense divider) and ~30 uF (settles in ~2 s).
Energize only after a stable reading inside the window; check VIN < 22 V first.

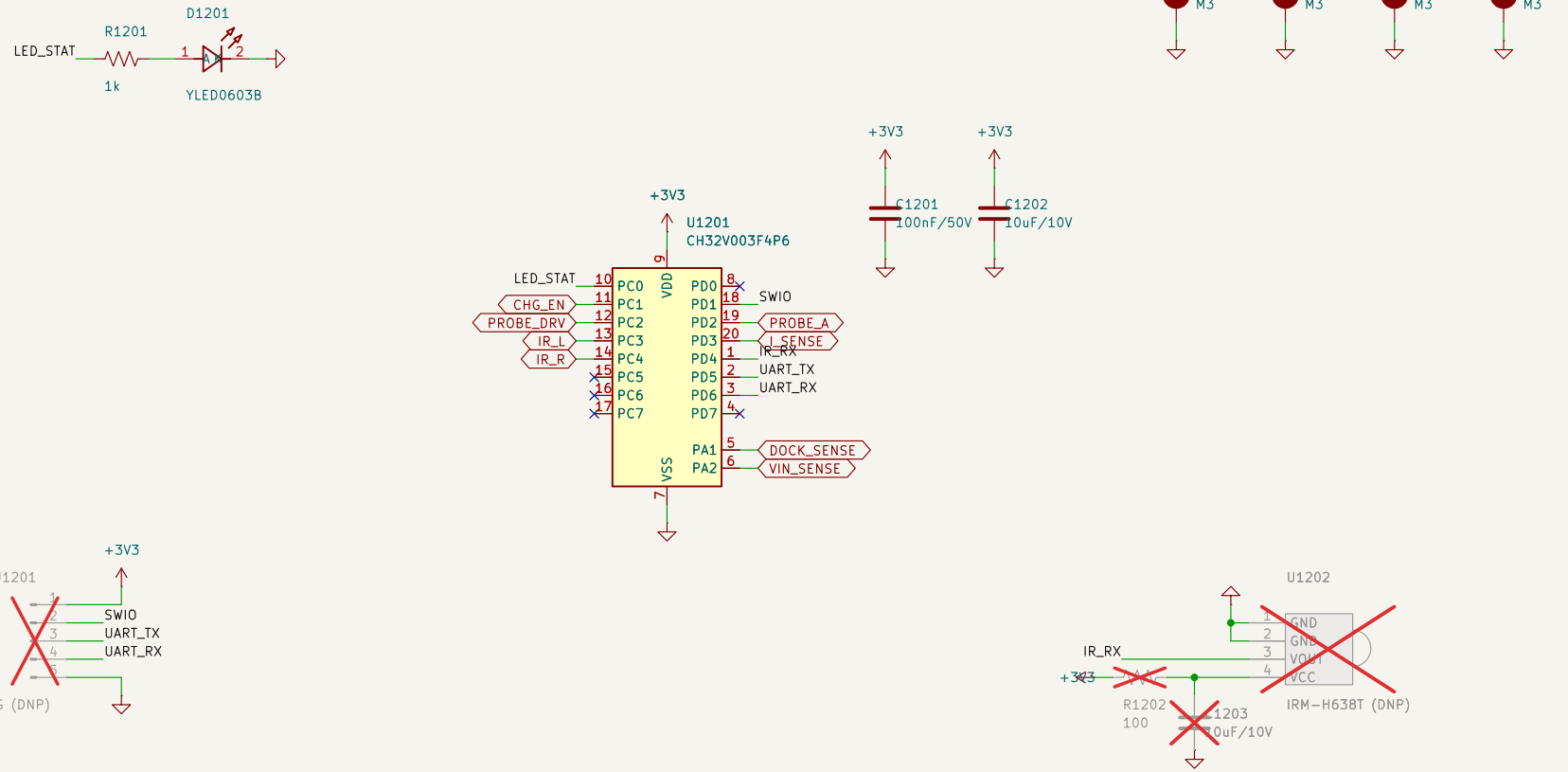
Q1001 soft start: R1004/R1005 set Vgs = −10 V, C1008 (Miller) ramps DOCK+ ~30 V/ms
(~0.5 A inrush into the robot input caps); turn-off ~3 ms.
Current: I_SENSE = 1 V/A (50 mOhm x 20), 3.3 A full scale. Turn off when it drops to ~0
(robot left) or on overcurrent (ADC analog watchdog). Route INA180 IN−/IN+ Kelvin to R1007.

**OOMWOO charging dock**

Sheet: /POWER/
File: Power.kicad_sch

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| Size: A4 | Date: 2026-10-27 | **Rev: 2** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 4/4 |




CH32V003F4P6: PA2 = VIN_SENSE (A0), PA1 = DOCK_SENSE (A1), PD2 = PROBE_A (A3), PD3 = I_SENSE (A4),
PC1 = CHG_EN (high = DOCK+ on), PC2 = PROBE_DRV, PC3/PC4 = IR_L/IR_R (TIM1_CH3/CH4, 38 kHz),
PC0 = status LED, PD4 = IR_RX (TIM2_CH1), PD5/PD6 = UART, PD1 = SWIO. HSI clock, no crystal.
ADC reference is VDD, and PROBE_DRV high is VDD too, so the probe math is ratiometric.

**OOMWOO charging dock**

Sheet: /CONTROL/
File: Control.kicad_sch

***Title: Control***

| Size: A4 | Date: 2026-10-27 | **Rev: 2** |
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| KiCad E.D.A. 10.0.4 | | Id: 5/4 |